Pricing and Relative-Value Dynamics of "Digital Asset Treasury" Equities: A Casestudy on Strategy (NASDAQ: MSTR)

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Outline

1. Introduction
2. Data and Structural Model
3. Calibration and Simulation
4. Results and Visualisation
5. Discussion

Motivation and Research Questions

- MicroStrategy (MSTR) has transformed into a large BTC-holding vehicle.
- Equity trades as a leveraged BTC proxy, but:
  - balance-sheet leverage,
  - BTC-per-share accretion,
  - and speculative equity premium

  are often conflated.
- **Research questions**
  - How much BTC growth (total and per share) is implied by current pricing?
  - How much leverage comes from the balance sheet vs. premium dynamics?
  - How severe are future funding needs and solvency risks?

Why These Indicators?

- **Implied BTC Growth Rate (IBGR)**
  - Measures expected BTC accumulation given current premium and financing behavior.
  - Total IBGR: growth of firm-wide BTC stack H_t.
  - Per-share IBGR: growth of BTC per share $B_t = H_t/N_t^{\text{sh}}$, i.e. accretion vs dilution.
- **Leverage and Elasticities**
  - ILE: pure balance-sheet BTC beta from A_t and D_t.
  - TEE: what equity traders actually feel once premium reacts to BTC.
- **Premium Mean-Reversion Index (PMRI)**
  - Z-score of current premium versus its OU long-run mean.
  - Answers: is MSTR rich or cheap vs its own history?
- **IFRD and Survival**
  - IFRD: distribution of dollar funding needed to finance future BTC purchases.
  - Survival: probability of avoiding a leverage/distress barrier.

Indicator Definitions I: Leverage and BTC Growth

Implied Leverage Elasticity (ILE).

- From the structural relation
$$E_t = e^{\pi_t}(H_t S_t - D_t)^+,$$
holding π_t fixed gives the structural BTC beta
$$\mathrm{ILE}_t = \beta_{LS}(t) = \left.\frac{\partial \log E_t}{\partial \log S_t}\right|_{\pi_t} \approx \frac{H_t S_t}{H_t S_t - D_t},$$
for $H_t S_t > D_t$.
- We compute ILE_t directly from the daily panel and report its time-series and current value.

Total Equity Elasticity (TEE).

- Premium reacts to BTC shocks:
$$\Delta\pi_t \approx a + \gamma_{\pi S}\,\Delta \log S_t + \varepsilon_t$$
(estimated by OLS on the panel).
- Total BTC beta combines structural leverage and premium response:
$$\mathrm{TEE}_t \approx \gamma_{\pi S} + \beta_{LS}(t).$$

Indicator Definitions II: Per-Share IBGR, PMRI, IFRD, Survival

Implied BTC Growth Rate (IBGR).

- Holdings dynamics: $dH_t = \alpha\pi_t^+ dt + dH_t^{\text{jump}}$.
- Calibrate α from non-event periods, and jump intensity λ_M and mean jump size $\mathbb{E}[\Delta H]$ from purchase history.
- At the current state (H_0, π_0), the short-horizon expected BTC growth rate is

$$\mu_H \approx \frac{1}{H_0}\left(\alpha\pi_0^+ + \lambda_M \mathbb{E}[\Delta H]\right).$$

Per-Share IBGR.

- BTC per share: $B_t = H_t / N_t^{\text{sh}}$.
- Closed form is messy because both H_t and N_t^{sh} can jump.
- We estimate the per-share growth rate by Monte Carlo:

$$\mu_{H/N} \approx \frac{1}{B_0}\mathbb{E}\left[\frac{B_T - B_0}{T}\right] \approx \frac{1}{B_0}\frac{1}{N_{\text{paths}}}\sum_{i=1}^{N_{\text{paths}}} \frac{B_T^{(i)} - B_0^{(i)}}{T}.$$

Indicator Definitions III: Per-Share IBGR, PMRI, IFRD, Survival

Premium Mean-Reversion Index (PMRI).

- OU parameters $(\kappa, \theta, \sigma_\pi)$ imply a stationary variance $\sigma_\pi^2/(2\kappa)$.
- PMRI is the z-score of the current premium:

$$\text{PMRI} = \frac{\pi_0 - \theta}{\sigma_\pi/\sqrt{2\kappa}}.$$

Implied Funding Requirement Distribution (IFRD).

- For each simulated path,

$$G_T^{(\text{path})} \approx \sum_k S_{t_k}\left(H_{t_{k+1}} - H_{t_k}\right)$$

  is the cumulative dollar cost of BTC purchases over horizon T.
- IFRD is the empirical distribution of $\{G_T^{(\text{path})}\}$: mean, variance, and quantiles (e.g. 5%, 50%, 95%).

Survival Probabilities.

- Define a distress barrier $H_t S_t \leq (1+\varepsilon) D_t$.
- For each path, record whether the barrier is ever hit before T.
- Survival probability is estimated as $\mathbb{P}(\tau > T) \approx \frac{\#\{\text{paths with no hit up to } T\}}{\#\{\text{all paths}\}}$, for $\varepsilon = 0$ (pure insolvency) and for a safety buffer, e.g. $\varepsilon = 0.1$.

Balance Sheet and Premium

- BTC price: S_t; holdings: H_t; debt: D_t; shares: N_t^{sh}.
- BTC asset value:
$$A_t = H_t S_t.$$
- Net asset value (limited liability):
$$\mathrm{NAV}_t = (A_t - D_t)^+ = \max(H_t S_t - D_t, 0).$$
- Equity market value:
$$E_t = P_t N_t^{\text{sh}}.$$
- BTC per share:
$$B_t = \frac{H_t}{N_t^{\text{sh}}}.$$
- Premium (for data work) with small floor ε:
$$\mathrm{NAV}_t^{\text{clip}} = \max(A_t - D_t, \varepsilon), \qquad \pi_t = \log \frac{E_t}{\mathrm{NAV}_t^{\text{clip}}}.$$

Dynamics under $\mathbb{P}$

- **BTC price**

$$\frac{dS_t}{S_t} = \mu_S \, dt + \sigma_S \, dW_t^{S,\mathbb{P}}.$$

- **Premium OU**

$$d\pi_t = \kappa(\theta - \pi_t) \, dt + \sigma_\pi \, dW_t^{\pi,\mathbb{P}},$$

with $\mathrm{corr}(dW_t^{S,\mathbb{P}}, dW_t^{\pi,\mathbb{P}}) = \rho$.

- **Holdings "flywheel"**

$$dH_t = \alpha \pi_t^+ \, dt + dH_t^{\text{jump}}, \qquad \pi_t^+ = \max(\pi_t, 0),$$

where jumps are BTC purchases identified from the purchase-history table.

- In our estimates:
  - accumulation is almost entirely jump-driven ($\alpha \approx 0$),
  - jump intensity and size are calibrated from actual BTC purchase events.

Calibration Results I: BTC and Premium

- Sample: 2021-01-04 to 2025-12-01, excluding tiny or negative NAV days.

BTC dynamics (under $\mathbb{P}$)	
μ_S	$\approx 14\%$ annual drift
σ_S	$\approx 49\%$ annual volatility
Premium OU	
κ	≈ 4.08 (fast mean reversion)
θ	≈ 0.71 (long-run log-premium)
σ_π	≈ 1.42

- BTC is very volatile but with moderate positive drift under $\mathbb{P}$.
- Premium mean reverts quickly to a slightly positive long-run level.

Calibration Results II: BTC–Premium Link and Holdings

BTC–premium link	
ρ	≈ -0.49 (negative correlation)
$\gamma_{\pi S}$	≈ -1.48 (premium falls when BTC jumps)
Holdings dynamics	
α	≈ 0 (no smooth premium-driven accumulation)
λ_M	≈ 17.3 BTC purchase events per year
$\mathbb{E}[\Delta H]$	$\approx 6{,}900$ BTC per event

- Equity premium tends to compress when BTC rallies sharply (risk-off response).
- BTC stack growth is dominated by lumpy debt/equity-funded purchase events.

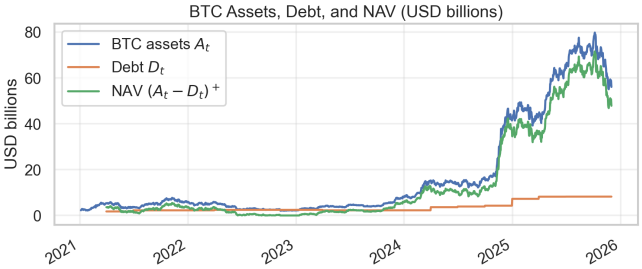
Simulation and Indicator Computation

- Simulate joint paths $(S_t, \pi_t, H_t, N_t^{\text{sh}}, D_t, E_t, B_t, \mathrm{NAV}_t)$ under $\mathbb{P}$ for $T = 3$ years, daily $\Delta t = 1/252$.
- Use Cholesky to generate correlated shocks (Z_S, Z_π).
- Holdings:
  - continuous term $\alpha \pi_t^+ dt$ (here negligible),
  - Poisson jumps with intensity λ_M and size distribution from purchase history.

Simulation and Indicator Computation (cont'd)

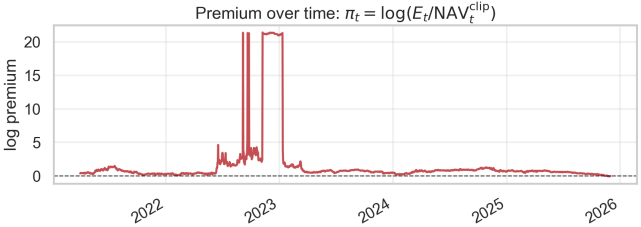
- From simulated paths compute:
  - **Total IBGR**:
$$\mu_H \approx \frac{1}{H_0}\left(\alpha\pi_0^+ + \lambda_M \mathbb{E}[\Delta H]\right).$$
  - **Per-share IBGR**:
$$\mu_{H/N} \approx \frac{1}{B_0}\mathbb{E}\left[\frac{B_T - B_0}{T}\right].$$
  - **PMRI**:
$$\text{PMRI} = \frac{\pi_0 - \theta}{\sigma_\pi/\sqrt{2\kappa}}.$$
  - **IFRD**: empirical distribution of
$$G_T \approx \sum_k S_{t_k}(H_{t_{k+1}} - H_{t_k}).$$
  - **Survival**: fraction of paths with $H_t S_t > (1+\varepsilon)D_t$ for all $t \leq T$.

BTC Assets, Debt, and NAV



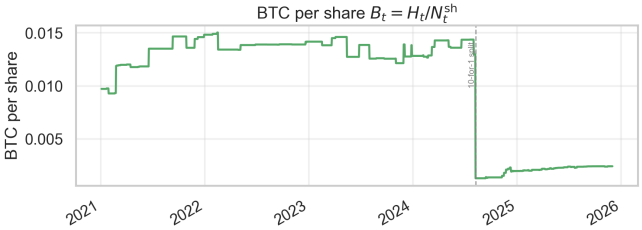
- BTC assets A_t, debt D_t, and NAV are shown in USD billions.
- Upward steps in A_t correspond to BTC purchase events; D_t jumps when new debt is issued.
- $\mathrm{NAV}_t = (A_t - D_t)^+$ highlights periods where the BTC stack barely covers liabilities.
- This plot is the foundation for leverage and survival analysis.

Premium over Time



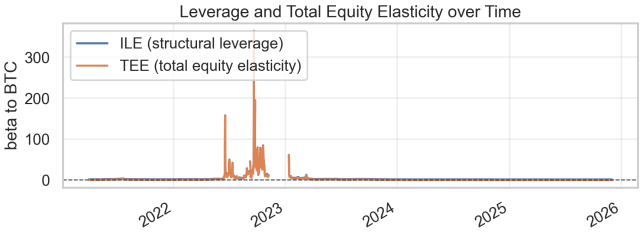
- Premium $\pi_t = \log\left(E_t/\mathrm{NAV}_t^{\mathrm{clip}}\right)$ measures how rich or cheap equity is vs. balance- sheet NAV.
- The series oscillates between deep discount and exuberant premium, with several distinct regimes.
- We fit an OU process to this series (after excluding tiny-NAV days) to obtain κ, θ, and σ_π.
- Current premium (2025-12-01) is slightly negative and about 1.5 standard deviations below the long-run OU mean (PMRI ≈ -1.5).

BTC per Share



- BTC per share $B_t = H_t/N_t^{\mathrm{sh}}$ shows how each share participates in the BTC stack.
- The sharp 10x drop in 2024 marks the 10-for-1 stock split (annotated in the figure).
- After the split, B_t trends upward again as new BTC purchases accumulate on a larger share base.
- Per-share IBGR from the Monte Carlo simulation summarizes the forward-looking BTC/share accretion rate implied by today's state.

Leverage and Total Equity Elasticity



- ILE (blue) is the structural BTC beta:

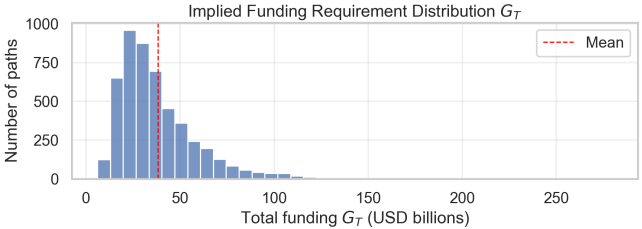
$$\beta_{LS}(t) \approx \frac{H_t S_t}{H_t S_t - D_t}.$$

- TEE (orange) adds premium sensitivity:

$$\beta_{\mathrm{TOT}}(t) \approx \gamma_{\pi S} + \beta_{LS}(t).$$

- Periods where TEE $\ll$ ILE indicate that premium tends to move opposite BTC (risk-off), dampening equity beta.
- As of 2025-12-01, ILE ≈ 1.17 while TEE ≈ -0.31, showing that sentiment dominates structural leverage in recent data.

Implied Funding Requirement Distribution



- Histogram shows the Monte Carlo distribution of

$$G_T \approx \sum_k S_{t_k} \left(H_{t_{k+1}} - H_{t_k} \right)$$

over a 3-year horizon, scaled to USD billions.

- Under current calibration:
  - mean $G_T \approx 3.8 \times 10^{10}$ USD,
  - 95th percentile $\approx 8.0 \times 10^{10}$ USD.
- Combined with survival probabilities, IFRD quantifies how aggressive continued BTC accumulation is under the historical measure.

Interpretation and Limitations

- MSTR currently trades near NAV with moderate structural BTC beta (ILE ≈ 1.2).
- Total equity elasticity is lower (even slightly negative) once premium response is included.
- BTC-per-share growth is strong but driven by lumpy financing events, not continuous premium-driven issuance.
- **Limitations**
  - Debt and share-count dynamics are deterministic in simulations.
  - BTC modeled as GBM under $\mathbb{P}$; no explicit jump or $\mathbb{Q}$ calibration.
  - Premium OU excludes jump regimes (crashes / manias).
- **Next steps**
  - Add issuance/conversion rules and net-debt dynamics.
  - Incorporate BTC option-implied volatility and jumps.
  - Explore scenario analysis for extreme BTC paths and funding constraints.

Thank You

Questions and discussion.